

IMPLEMENTATION DETAILS & SNIPPETS

Developer: [Left Blank] | OptiCrop Project Documentation

Repository: <https://github.com/goutham-05-raj/OptiCrop-Smart-Agricultural-Production-Optimization-Engine>

1. Backend Routing Code (Flask)

The controller logic utilizes Flask routes to accept POST data, process it, and render the results template. Below is an excerpt of the prediction pipeline:

- Extract soil parameters from the HTML form request.
- Transform the input list using the StandardScaler pipeline.
- Perform ensemble predictions using Random Forest classifier.
- Predict estimated yield and categorise soil with the KMeans model.
- Append predicted values to user_predictions.csv and render results.

OptiCrop - Key Code Modules

```
load_all_models() [app.py]
#Serialises best_model.pkl, yield_model.pkl, scaler.pkl,
#Deserialises best_model.pkl, yield_model.pkl, scaler.pkl
kmeans_model.pkl on server start into a shared dict.
predict [POST, app.py]
1. Parse & validate 7 form fields (bounds check)
/predict [POST, app.py]
/predict [POST, app.py]
```

2. Model Training Script (train_models.py)

The training pipeline uses StandardScaler to transform, splits them (80/20 train/test split), fits the StandardScaler, trains the classifiers, outputs charts, and serializes files using pickle.

```
1. Random Forest Classifier (100000 samples)
2. RF regress -> yield (t/ha)
3. RF regress -> crop label
4. RF regress -> fertilizer advice
5. KMeans Predict soil cluster
6. NPK delta -> fertiliser advice
7. Append row to user_predictions.csv
8. Append row to user_predictions.csv
9. Random Forest Classifier
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train_models.py
Scales, encodes, trains 4 classifiers, selects best by accuracy,
dumps pipelines to models/ directory.
```